

VARUN RISHWANDH SEKAR

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EDUCATION

NORTHEASTERN UNIVERSITY

Master's in Computer Science

Sept 2024 - June 2026

4.00 / 4 GPA

Teaching Assistant for Object Oriented Design (Spring & Summer 2025) and Program Design & Implementation (Fall 2025)

ANNA UNIVERSITY

Bachelor's in Computer Science and Engineering

June 2019 - April 2023

3.84 / 4 GPA

Gold Medal & Best Environmental Project, iGEM Synthetic Biology Competition 2022

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, Go, C/C++, Bash

Frameworks: React, Node.js, FastAPI, Spring Boot, Django, Flask, GraphQL, gRPC, LangChain

Databases: PostgreSQL, MySQL, MongoDB, BigQuery, Firebase

Infrastructure: AWS, GCP, Docker, Kubernetes, Terraform, Airflow, Nginx, CI/CD, Git

WORK EXPERIENCE

Software Engineer Co-op - Dell Technologies

Jan 2026 - June 2026

- Building the document ingestion backbone for an Agentic AI platform, enabling RAG-powered market research through automated extraction, summarization, chunking, and embedding stages across financial datasets.
- Architected a modular extraction framework for 7 data sources, replacing brute-force document scanning with targeted rule-based extractors in a plug-and-play architecture with legacy fallback for backward compatibility.
- Consolidated 8 sequential Airflow DAGs into a parent-child architecture with parallel execution, cutting pipeline redundancy and improving orchestration throughput across the entire ingestion layer.

Software Engineer - MR. COOPER

Nov 2022 - August 2024

- Designed and shipped core backend services (rule engine, matching service, data APIs) for a loan audit platform that automated document verification using semantic similarity, cutting manual processing from weeks to hours.
- Architected a plugin-based system for dynamic document type configuration via database, eliminating 20+ hours/month of developer effort and reducing deployment cycles from days to minutes.
- Integrated LLMs for automated document annotation with human-in-the-loop verification controls, reducing dataset curation time by 70% while maintaining financial compliance standards.
- Drove test coverage from 0% to 70% across production services handling millions of documents monthly. Recognized as "Rookie of the Year" out of 100+ new hires.

Java Development Intern - ZIROH LABS

Jan 2022 - March 2022

- Developed a Java application using Wikimedia API with a caching layer (reduced API calls by 60%) and AES-256 encryption for GDPR compliance, serving 500+ daily users.

Backend Developer Intern - GREY SCIENTIFIC LABS

Aug 2021 - Dec 2021

- Built RESTful APIs using Django Rest Framework for a hospital platform enabling 500+ doctors to view patient records and annotate medical images, with passwordless JWT/OTP authentication and 99.9% uptime.

PROJECTS

[AudioSeek](#) | 2nd Place, MLOps Expo @ Google Cambridge [Python, FastAPI, React, GCP, Docker, K8s]

- Built a full-stack application with a streaming chat interface, audio playback integration, and "time-travel" querying that lets users search specific timestamps across 500+ hours of audiobook content.
- Developed backend APIs with FastAPI handling intent classification, chapter boundary metadata, and secure session management supporting 1000+ concurrent users.
- Deployed on GCP using Cloud Run for the frontend, GKE for compute-intensive services, and GCS for storage, delivering real-time Q&A with synchronized audio playback.

[Distributed AI Inference System](#) [Go, AWS ECS, SQS, DynamoDB, Docker, k6]

- Designed an event-driven inference platform processing 5000+ concurrent ML jobs using Go microservices on ECS Fargate with auto-scaling tied to SQS queue depth and fault-tolerant worker pools with dead letter queues.
- Ran scalability experiments with k6/Locust demonstrating linear scaling to 16 workers, sub-second p95 latency under 200 req/s, and automatic recovery from 50% worker failure with zero message loss.